Arriving at the centre of consciousness: a transjective framework

Paper I of the Peace Papers, six papers on one framework; each stands alone. A reflective essay. Website version, working draft v17, 2 November 2026. Devon Summerhayes, London.

All figures are original diagrams, drawn or redrawn with the help of generative AI; none is reproduced from another publication. Supplementary Material (Part A, A1 to A4; Part B, Figures S1 to S13) accompanies this paper; it is now part 2 of the Unifying Paper.

Abstract

Scientific and spiritual accounts of experience are usually treated as rivals. This essay proposes a framework in which they are two views of one experience, and a way to the centre from which both are seen. It makes four contributions. First, a labelling discipline: every descriptive claim is marked Experienced, Measured or Interpretive, so that the two languages can be held together without either undermining the other; working that way is called scientuality. Second, three perspectives of the I: the felt, subjective perspective (I1, a spectrum with negative, neutral and positive settings), the understanding, objective perspective (I2), and the witness's eye (I3), through which both are seen together as one transjective view, from the stance the essay calls transjectuality. Three words are kept apart: the self is the looker, the witness is the self looking at its own perspectives, and I3 is the eye it looks through. Third, a picture, offered as Interpretive: I3 as a lens through which universal consciousness, ɪɪ, a mark and not a name, passes to its focal point, the singularity of consciousness, ɪɪ brought to a point in one person; neither is a higher level of reality. The essay sets the perspectives beside older maps of the mind, McGilchrist's divided brain and the three-part models of Freud, Berne and Jung, as parallels and not proofs. Fourth, an experiential test: the reader rests awareness at the centre of a drawing and reports whether feeling and understanding hold together, against a control, with a decentering score and thresholds set in advance. It is offered to be tried, not as a finding. Objections and the framework's own risks are answered in turn.

In the series, this is the first paper and the source of the framework the others use. Paper II applies the perspectives to empathy and compassion; paper III argues for the stance as the dissolution of paradox; paper IV carries the perspectives into the norms families and cultures pass on; paper VI reads artificial intelligence through them; and the Unifying Paper states the framework once and gives the cosmology in full (part 2).

1. Introduction

This essay proposes a transjective framework for arriving at the centre of consciousness. It tries to hold two things together: a measurable account of how biology and environment shape emotion and behaviour, and an older, felt account of the same human experience in the language of spirit, conscience and soul. Put in one document without a rule for telling them apart, each undermines the other. A scientist could not tell which statements were findings, and a spiritual reader could not tell which were metaphors. This essay keeps the ambition and fixes the structure: it treats the two languages as two views, and says clearly which view each statement is in. A third, integrating stance holds both views and dismisses neither; this essay calls the stance transjectuality, and the practice of working from it, with both languages held, scientuality. This essay sets out the framework. Its use for empathy and compassion, with four testable hypotheses, is the subject of a companion essay, "Empathy, compassion and the humanity paradox: a transjective account" (paper II), referred to below as the companion essay (where this essay relies on it, the relevant content is restated here). Table 1 sets out the three modes of claim and Table 2 aligns older three-part maps of the mind.

Notation (author's mapping, Interpretive). The letter I stands for the person's first-person self: the one who feels, understands and sees. Three perspectives are taken from it, labelled I1, I2 and I3. A

perspective is a place from which experience is taken up: it is not a part of the mind, a level of reality or a type of person. I1 is the subjective perspective, experience felt from the inside: the felt inner life, tied here to emotional empathy, and called spiritual in the sense of inner experience, not a separate substance. I2 is the objective perspective, experience taken as an object and understood from outside: analytic thought, tied here to cognitive empathy and executive function; "objective" here means understood as an object, not independent of any mind, and I2 is neutral, a tool that can serve any setting of I1. I3 is the transjective perspective, the eye of the witness: not a third, outside observer but the eye through which the felt and the understood are seen together and neither is dismissed, from the impartial stance the essay calls transjectuality; the witness is close to Adam Smith's impartial spectator (Smith 1759), I3 its eye, and from that stance care can extend beyond the people in front of us. It is not a further part of the mind and not a higher level of reality, and the witness does not act. Three words are kept apart in section 5: the self, the witness and I3, the witness's eye. In plain words the three are the heart (I1), the mind (I2) and the eye of consciousness (I3): emotional empathy is felt in the heart, cognitive empathy considered in the mind, and compassion is the heart directed by the mind, seen by consciousness when witnessed; these are perspectives, not parts of a person. Compassion is felt concern (I1+) directed by understanding (I2) and acted on, which is why it is a practice as well as a way of seeing; witnessed compassion is the same with the witness looking, and the witness's own part is to notice which state one is in, which is what makes moving between states possible. The soul, in this essay, is the whole person acting on both with the witness present, so "spiritual" (I1) and "soul" are different things here: the first is the felt inner life, the second is that life joined with understanding, witnessed and put into action. This essay is written from I3; working from it, with both languages held, is what it calls scientuality. Figure 1 draws I1 and I2 as the corners of a triangle, with a circle through them: I1− and I1+ at the ends of the base, I2 at the apex and I3, the witness's eye, at the centre of the circle, drawn as a ring with a point of light within it. The ring, the point and the circle are read fully in the Discussion (section 8); the test in section 6 does not depend on the reading. The Supplementary Material, now part 2 of the Unifying Paper, adds further images of I3 (the joining edges and two spirals, A1; the lens of the laser, A2; the lens of the eye, A3), not further meanings of the word: they differ as the projections of one shape differ, which is the point of the tesseract in A1, and a lens, joining edges and a meeting point are how one perspective looks when it is pictured from different sides.

I1 is a spectrum, from the negative pole (I1−) through neutral to the positive pole (I1+), as the base of Figure 1 draws it; its three settings are regions of that range, not separate states. I1 stands at one of three settings: positive (I1+: conscience, trust, grace, felt connection), neutral (calm, unengaged feeling) or negative (I1−: the self-directed pole, in two forms: reactive, as fear and being overwhelmed, and appetitive, as appetite and instrumental aggression; in both, felt concern for others is absent). The plus and minus mark polarity, like the poles of a magnet, not good and bad: each has a job, and the problem is one governing alone. I1+ tends to support prosocial action when I2 directs it, but conscience can also be rigid and felt connection narrow; I1− tends toward self-protection, which can become antisocial when it governs alone or when I2 serves it without felt concern (Interpretive; a state, not a type of person). Neutral is calm with nothing asked of it and no concern stirred, and at its dead end it is dullness, calm because nothing is alive in it, what meditation manuals call sinking and the ingredient of indifference. Love at rest, calm with concern present and nothing asked of it, equanimity, is not neutral but I1+ at rest; its wakefulness resembles the tonic alertness Metzinger (2020) places at the core of minimal phenomenal experience, which on his account is nearly contentless, so the resemblance is partial (Interpretive). The quiet middle is also where thought sees clearest: fear scatters the light and appetite pulls it in, and at neutral nothing bends it, which is why, on this reading, the contemplative order is calm first and insight after, the attentional family of practice before the deconstructive (Interpretive; the families are those of Dahl, Lutz and Davidson 2015). I2 plays one of three roles: offline (I1 "alone"), present while the feeling sets the course (I1 "with" I2), or setting the course (I1 "directed by" I2). I2 has no sign of its own, so each combination takes its sign from I1; the two forms of I1− are forms of one setting, not a fourth setting, so the count stays at nine. Six states are named, in five of the nine combinations, one of them twice over by the form of I1− (the companion essay sets them out in a table: paper II, section 6, Table 3): I1+ directed by I2 is compassion, neutral I1 with I2 is indifference, I1− in its reactive form with I2 is

defended detachment, I1− in its appetitive form directed by I2 is cruelty, I1− in its reactive form directed by I2 is revenge, and I1− in its reactive form alone is distress. Each combination can be witnessed (I3 taken up, the witness looking through it) or not; compassion, as used here, is the first combination, and witnessed compassion is that combination with I3 taken up. Section 5 sets the perspectives beside older maps of the mind and says what the witness rests on.

Position and method

The framework comes mainly from the author's self-examination and from informal observation and interpretation of other people over many years, checked afterwards against published research. It is therefore observational and introspective in origin, not clinical or experimental. These are limited sources: people can be wrong about their own minds and others', and unsystematic observation is open to bias. The framework is offered as a hypothesis to test, not a finding. No identifiable individual is described and no case material is reported. Measured claims are drawn from published sources and marked as such. Experienced and Interpretive claims are the author's own. A framework of this kind is a map, not the territory, and a map can become a hiding place: categories can be a way of avoiding the harder, personal parts of life that the Experienced mode points to. The three modes are meant to guard against this by keeping the felt account in view beside the measured one, but only practice can show whether they do.

How the framework was developed and checked. The scheme was drafted from the author's observations and revised as inconsistencies were found. No systematic literature search was made: sources were chosen to test and sharpen the framework, so they may lean towards support, and were not appraised for quality. Numerical claims were checked against the original papers or their abstracts, several against abstracts only. Generative AI helped draft and edit the text, checked consistency and references, rendered the figures and contributed design details of the experiential test (see Declarations).

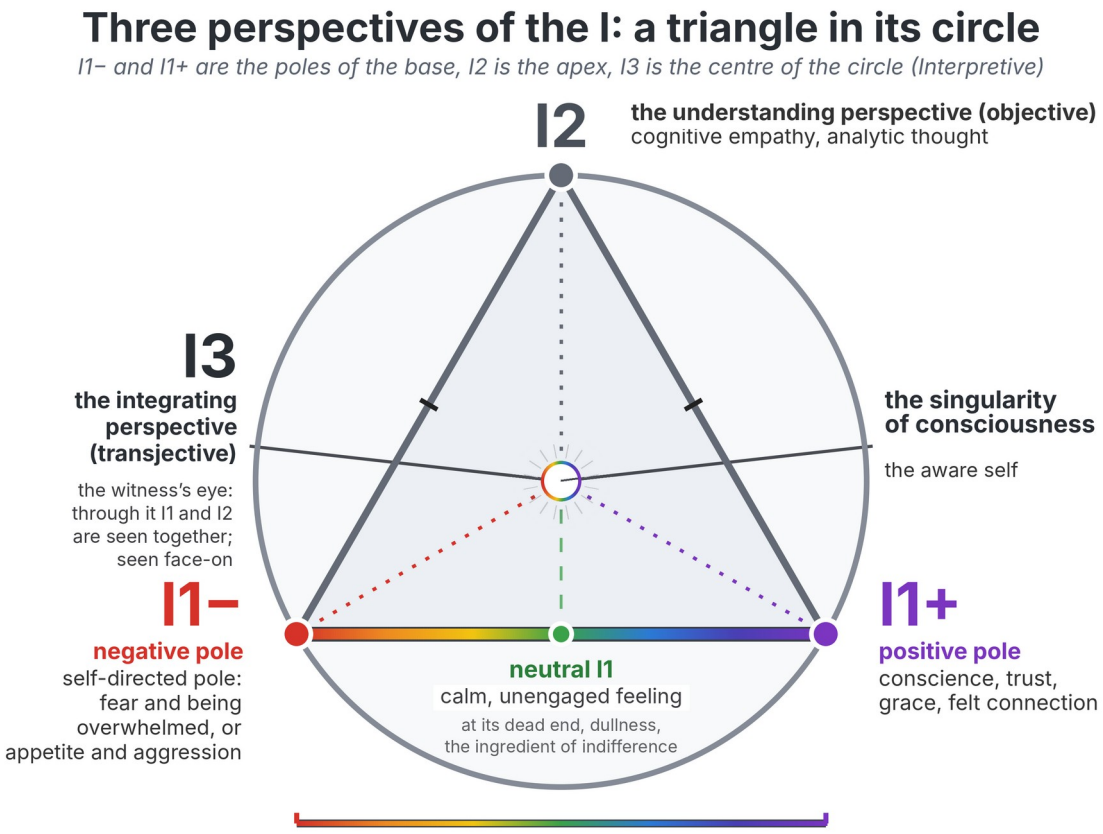


Figure 1. A map of the perspectives of the I, with the centre of consciousness at its middle (Interpretive). I1– and I1+ are the poles of the felt perspective (subjective), at the ends of the base, neutral I1 between them, the negative pole labelled with both forms (Introduction). I2, the understanding perspective (objective), is the apex, equally far from both poles. I3, the witness's eye (transjective), is the centre, equally near each corner, drawn as a ring: the eye of the witness face-on, with faint rays for light given. The point of light within it is the singularity of consciousness, ɪɪ brought to a point at the eye's focal point: the self, the looker, the source of the light given; the circle through the three corners is ɪɪ seen whole. Lines join I3 to each perspective. Emotional empathy sits at I1 and cognitive empathy at I2; compassion combines the two (Introduction) and has no position on the figure. Colour: the spectrum, red at I1– through green to violet at I1+, is I1, carried by the ring because the whole felt range passes through the witness's eye; grey is I2; white is ɪɪ. Green marks only neutral's place on the scale. A picture of how the perspectives relate, not a measurement.

2. What "transjective" means here

A transjective view is neither purely subjective nor purely objective. Meaning arises in the relation between a knower and what is known. The shade of a tree is not only in the tree and not only in the person standing in it. In this essay, "transjective" is used in that plain sense: some of the most important things about human life only show up when the inner view and the outer view are held together and neither is dismissed.

The empathy spectrum is a worked example. Where a scientist sees a trait that varies across people and is measured by questionnaire and by the brain's response to another's pain (Singer et al. 2004), another person sees conscience, heart or soul. These are not rival claims. They are the same thing described from outside and inside (Interpretive). The empathy spectrum runs across people; the I1 spectrum of the Introduction runs within a person over time, and the two are different things.

Related work and what is new

Relating science and religion in more than one way is not new. Barbour (1997) described four: conflict, independence, dialogue and integration. Gould (1997) proposed non-overlapping magisteria, a form of independence. This essay sits nearer dialogue and integration, but it does not ask religion to settle empirical questions or science to settle questions of meaning. Varela (1996) argued that first-person accounts and brain data should constrain each other (neurophenomenology). Dual-aspect monism, a family of views with roots in the Pauli-Jung correspondence and developed by Atmanspacher and Rickles (2022), treats mind and matter as two aspects of one underlying reality. The word "transjective" is rare. In philosophy it is used for what is neither purely subjective nor purely objective; this essay found no single authoritative origin for it, although John Vervaeke uses it in his lectures, proposing it as a term for the relation between agent and world that is neither only subject nor only object (Vervaeke 2019, episode 31). It is used here only in the plain sense given above, and the essay does not claim to have coined it; the noun transjection, for the operation of holding a pair as one, is the essay's own and is defined where it is used (section 5). The words "transjectuality", used here for the stance taken through I3 and built on "transjective", and "scientuality", used for the practice of holding the two languages together, are coined here.

Within psychology, the felt and understood perspectives have close relatives. Empathy research separates affective from cognitive components (Decety and Jackson 2004; Zaki and Ochsner 2012) and traces them to perception-action mechanisms with deep evolutionary roots (Preston and de Waal 2002); the companion essay takes up that literature, and the work on compassion, in detail (paper II, section 2). Dual-process theories contrast fast, autonomous processing with slower, reflective processing (Evans and Stanovich 2013); that pairing resembles the instinctive layer of Table 2 and I2 only loosely, as the Freudian maps do.

Closer still are three literatures the witness touches directly. In clinical psychology, decentering, observing one's own thoughts and feelings as passing events rather than as oneself, is a reviewed metacognitive process with measurable effects (Bernstein et al. 2015), and the contemplative-science account of meditation distinguishes attentional, constructive and deconstructive families of practice (Dahl, Lutz and Davidson 2015); I3 taken up is close to decentering, and the bliss states of the Supplementary Material (the Unifying Paper, part 2, A3) to the deconstructive family. In phenomenology, Zahavi (2005) argues that a minimal, pre-reflective self-awareness accompanies every experience; the witness of this essay is that awareness taken up reflectively (section 5). In the study of "pure awareness", Metzinger (2020) proposes minimal phenomenal experience as a target for description and modelling, and Josipovic (2014) reports neural correlates of nondual awareness in meditators; the essay's π is offered beside these as an image, not a competing account. None supplies the labelling discipline that lets the felt and the measured descriptions of one position sit in one text. The nearest precedent is the distinction between first-, second- and third-person data in the first-person methodologies of Varela and Shear (1999); what the three modes add is the Interpretive mode, for readings that are neither report nor measurement, and a stated standard of judgement for each mode. The lens image of the Discussion is placed beside its own relatives there (section 8).

What this essay adds is not a new mechanism. Its first claim is a labelling discipline, the three modes of claim, applied to a domain where felt and measured statements are usually mixed. Its second is an arrangement: three perspectives of the I, with the witness's eye, I3, at the centre, the felt perspective set between two poles and a neutral setting, and the three settings crossed with the three roles of understanding to give nine combinations, with six states named in five of them. Its third is an experiential test of the witness that a reader can try (section 6), with a picture, read fully in section 8, of the witness's eye as a lens and the point of light at its centre as the singularity of consciousness, π brought to a point. The empathy and compassion hypotheses are in the companion essay (paper II, section 7).

The first two are practical and make no claim about what awareness is. The third carries a cosmology, one light in many lenses, and the essay says so: it is offered in the Interpretive mode,

judged by coherence and usefulness, placed beside its neighbours in section 8, and never advanced as a finding; the essay does not claim to resolve the mind-matter problem.

3. The discipline: three modes of claim

The bridge only holds if every descriptive statement declares its mode. This is the rule the essay follows, and Table 1 sets out the three modes. Definitions (what a word means here) and statements of value are not claims about the world, so they sit outside the modes; the one value commitment in this work (the Unifying Paper, part 2, A3) is marked as such.

Table 1. The three modes of claim.

Mode	Example	How to judge it
Experienced	A sense of stillness, of being watched by something larger, of a spiral up or down.	By honesty and consistency with other accounts. It is testimony, not proof, and need not be defended as physics.
Measured	A study of 1,011 people found no evidence that individuals are "left-brained" or "right-brained" (Nielsen et al. 2013).	By evidence, replication and effect size.
Interpretive	Freud's Id, Berne's Child and the "chimp" are three descriptions of one instinctive layer of the mind.	By coherence and usefulness: does it clarify more than it distorts?

Two consequences follow. First, words like "proves" belong only to Measured claims. Second, physics and mathematics may be used as images in Experienced or Interpretive claims (the spirals and lattices of the Supplementary Material are examples), but not as evidence for souls.

A worked example

A friend tells you that her father has died. The same moment can be described in three modes.

- **Experienced:** a tightness in the chest, and a wish to sit with her.
- **Measured:** watching or hearing about another person's distress engages some of the same affective brain regions as one's own distress (Singer et al. 2004, for pain). This says nothing about what the tightness means.
- **Interpretive:** the tightness is a form of caring, and acting on it with an understanding of what she needs is what this essay calls compassion.

Each statement is judged by its own standard (Table 1). Every word for a feeling is a projection of it onto the plane of understanding, a shadow with fewer dimensions than the feeling, which is why the Experienced mode is judged by honesty and consistency rather than by evidence, and why traditions can name one felt range in different words without contradiction. Problems begin when one is offered in the voice of another, for example when the interpretive reading is presented as a brain-imaging result.

4. A parallel: the divided brain (Interpretive)

McGilchrist's The Master and His Emissary (2009) is the closest published parallel to the I1, I2 and I3 pattern, so it is used here to ask whether the pattern is plausible, not to prove it. His thesis is that the two hemispheres are best understood as two modes of attention, not as "logical" and "creative" halves, which he rejects: the left attends narrowly, to detail and to what can be grasped and manipulated; the right attends broadly, to the whole, to context, to the living and the relational. Read beside the perspectives (author's mapping, Interpretive), I1 resembles the broad mode, I2 the narrow mode, and I3 the movement McGilchrist describes, in which the broad mode meets the world first, the narrow mode analyses, and the broad mode takes the result back into the whole; that cycle is a summary of the book's argument, not a quotation. The brain basis is contested. The popular story that individuals are "left-brained" or "right-brained" has no support (Measured; a study of 1,011

people; Nielsen et al. 2013), and McGilchrist does not make that claim; his claim, that the hemispheres differ in style of attention, has been criticised as over-interpreting what is known about brain function in order to explain cultural change (Gibbins 2010); and lesion work shows a double dissociation between the two empathies by prefrontal region, inferior frontal for the emotional and ventromedial for the cognitive, and did not test a left-right split (Measured; Shamay-Tsoory et al. 2009). The resemblance is therefore a loose analogy about styles of attention, used as a metaphor and not as evidence about the brain.

5. Parallel maps of the mind and the perspectives (Interpretive)

Why this section: several older maps of the mind already divide it into layers that resemble felt, reasoning and integrating functions. Lining them up shows that I1, I2 and I3 loosely parallel an old pattern, not a new discovery. It also says what the witness rests on (the note on the witness below).

Several three-part models of the mind can be set side by side (Freud 1923; Berne 1964; Jung 1966; Peters 2012). They are aligned here by function (Table 2). They are loose correspondences, not identities: Jung's persona, for instance, is not the same as Freud's superego. Berne's three ego states also give the felt perspective a second vocabulary, since each has a felt side: I1 can be occupied as Child (feeling raw: fear, appetite, delight), as Parent (feeling as an introjected rule: care in its nurturing form, aggression in its critical one) or as Adult (feeling observed calmly). Set against the settings, in the functional names later transactional analysis gave Berne's states (his own were the natural and the adapted Child), I1+ resembles the nurturing Parent and the free Child, the reactive form of I1− the adapted Child, the appetitive form the free Child's appetite and the critical Parent's aggression, and neutral the Adult's feeling (author's mapping, Interpretive; loose, as the table is).

Table 2. Three-part models of the mind, aligned by function (Interpretive).

Role	Freud	Berne	Jung	Peters (Chimp Paradox)
Instinctive, self-directed layer (I1−)	Id	Child	Shadow (wider than the Id)	Chimp
Mediating, reality-testing self (the ego: I1 and I2 in the states)	Ego	Adult	Ego	Human
Learned rules and social self (the superego: I2's rules with I1's force)	Superego	Parent	Persona	Computer

The perspectives beside Freud's structures (author's mapping, Interpretive). I1− in its appetitive form is nearest the Id; its reactive form, fear, is not, since Freud placed anxiety in the ego ("the actual seat of anxiety"; Freud 1923), so the instinctive layer of Table 2 and the Id overlap only in part. I1+ is conscience, drawn here apart from the superego (rules taken in from society), which departs from Freud, for whom conscience is a function of the superego; plus grace, trust and felt connection. I2 is the ego's analytic, reality-testing thinking (what Freud calls secondary process), close to Berne's Adult, a state built from I2 with neutral I1; the ego itself is not a perspective and not the aware self. Read this way, Freud's structures are mixtures of the perspectives, the id I1− in its appetitive form alone, the ego I1 with I2, the superego I2's rules with I1's force: the perspectives cut across the structures, which answers the objection that the scheme only relabels, since a relabelling keeps the level and this takes the structures apart, though it does not show that the perspectives are the deeper layer. Jung's Shadow is wider than the Id, taking in everything a person disowns, so it matches the instinctive layer in part; Jung's Self, the whole that holds conscious and unconscious together, is a parallel to the self, the centre and the whole at once. Figure 2 shows compassion as the overlap of I1 and I2, the seam, and I3 as the eye from which the seam is seen, drawn above it on the page only.

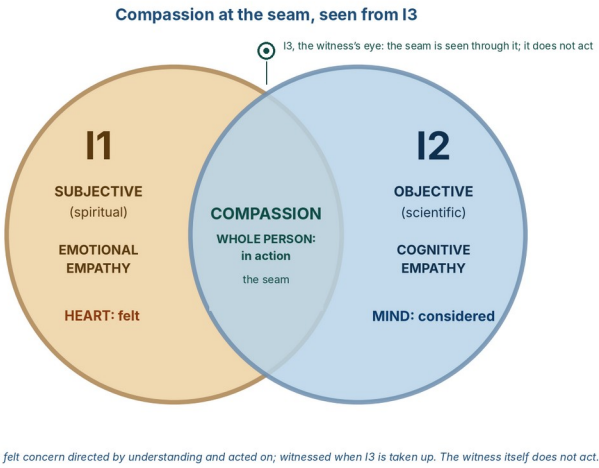


Figure 2. The two empathies and their seam (Interpretive). I1, the felt perspective (subjective, felt and spiritual), holds emotional empathy (heart); I2, the understanding perspective (objective, understanding and scientific), holds cognitive empathy (mind); compassion, felt concern directed by understanding and acted on, is their overlap, the seam. I3, the witness's eye (transjective), is drawn above it as the eye from which the seam is seen; above on the page only, since the eye is not a higher level of reality, and it does not act: the person acts from the seam. The overlap is drawn for the positive pole of I1; with I1− the same region holds cruelty and revenge, and defended detachment beside it (Introduction).

Used as descriptions of states, the perspectives are a useful lens: most people move between them (Interpretive; H3 in the companion essay (paper II, section 7) tests the nearest measurable version, whether a person's emphasis on feeling or on thinking behaves as a state or as a trait), and people differ in how easily they can step back and observe their own reactions. Where the aware self is not yet looking, which is to say where I3 is not yet taken up, the person is simply in the state. In a mind racing with worry, the fear is subjective (I1−) and the thoughts that go with it are I2, which may know better without steering: I1− in its reactive form with I2, but not directed (Interpretive). The companion essay calls this combination defended detachment, whose usual form is numbing and avoidance; worry is the same combination in a racing form, since the scheme carries no measure of arousal and one combination holds numbing and racing alike. Used as a way to sort people into kinds, the scheme fails (Interpretive; the same H3), and people ascend and descend over a lifetime. This essay keeps the first use and drops the second.

A note on the witness (author's mapping, Interpretive). The witness, as this essay uses the word, is reflective: the I turning to look at its own perspectives from within, through I3, which a person does intermittently and can learn to do more often. It rests on something that is not intermittent: the minimal, pre-reflective self-awareness that phenomenologists hold accompanies every experience and is not itself an act of observation (Zahavi 2005). In Figure 1 that constant self-awareness is the point of light at the centre, the aware self, and the ring around it, the witness's eye, is always there too; I3 taken up is that eye looked through reflectively, which is intermittent. The witness of this essay is therefore that constant awareness taken up reflectively, not a second observer added to it, which is why, when the I turns on itself, witness and witnessed are the same point of light, and nothing further is needed to witness the witness. Three words are kept apart here. The self is the looker and the source of the light, the constant awareness at the centre: the centre of the title. Being π emanated, the self can merge with the all, the source returning to the light it is of, and can come back: the in-breath and out-breath of the Supplementary Material (the Unifying Paper, part 2, A3). The witness is the self looking at its own perspectives. I1 and I2 are perspectives whose settings are states and can be measured; I3 is the eye they are seen through; and the witness is the self looking, the self being the I of the Notation from which the three are taken. I3 is the eye of the witness, the eye the self looks through: the eye and not the looker. The looker cannot observe itself; it sees of itself only its perspectives, reflected in I3, and is not seen but known, aware that it is aware, and that knowing is the pre-reflective self-awareness above. The self and the witness are transjections of each other, a transjection being a pair held as one from the stance that sees both: the

looker and the looking, one point of light from two sides. A paradox, on this account, is two perspectives taken for one (argued in paper III, section 1). The witness and the void hold each other: the void is the opening the light comes through and the witness the looking through it, which is why the point at the centre of Figure 1 is white, of no single colour; what the eye surrounds is not a thing to be seen but the opening seeing happens through. Source and opening are one point from two sides: looked at from the states, the point is where the light comes from; looked from, it is the opening the light passes through, and nothing is found there because the looking is there (Interpretive). From inside, the opening is felt as hollow: an emptiness one looks from, not at (Experienced). The void is empty so that one can shine one's light into it, focus it as a state narrows it and, if one chooses, feel it; the now is the presence that results, the stance from which the states are seen (Experienced and Interpretive). I3 is the eye of the witness, how it is focused: taken up, it converges on feeling and understanding together, and from there I1 and I2 appear as states a person is in rather than places looked from. Taking up I3 is not dissociation. The defended row of the companion essay is feeling walled off from understanding; from I3 both are present and seen together, so refocusing through I3 re-associates what the state had split, scattered light gathered again to a point, which on this reading is the difference between decentering and detachment (for decentering, Bernstein et al. 2015). What I3 taken up provides is flexibility: staying in contact with what is felt and thought and still choosing, which acceptance and commitment therapy calls psychological flexibility and roots in an observing self, self-as-context, the witness under another name (Hayes et al. 2006); Schooler's (2002) meta-awareness is the nearest construct in consciousness science. In one more image, from the book section 4 drew on (McGilchrist 2009), the self is the master and the ego, the I as it acts in its states, is the emissary: sent into the states to act and return, it serves only while it remembers it is not the master, and identifies with a state when it forgets, which is the essay's closing advice in one picture; the witness is how the self keeps its emissary in view. Only the relation is borrowed; the hemispheres stay where section 4 put them. Jung's Self (above) is a parallel to the self and the impartial spectator (Introduction) to the witness; neither is an extra definition; this is an Experienced and Interpretive reading, not a measured claim.

A possible link to Panksepp's model (Interpretive). In the neuroscience of affect, primary processes are raw feelings, secondary processes are learning and memory, and tertiary processes are the higher cognitive forms of consciousness built on them (Solms and Panksepp 2012). Tertiary processes are cognition, so they resemble I2, not I3, and "higher" here means higher in the brain, not better. Reflecting on one's own feelings may nonetheless be a capacity that the witness needs. If so, tertiary-process reflection could be a precondition of I3 without being I3 itself. This is a suggestion, not a finding.

The three triads, side by side. Table 3 sets the essay's three triads, two languages, three modes of claim and three perspectives, beside each other; the rows line up loosely, not as equivalences, since a claim from any perspective can be made in any mode, and the modes classify claims, not people or places.

Table 3. The three triads the essay uses, side by side (Interpretive).

Triad	First	Second	Third
Languages	Spiritual: the felt account	Scientific: the measured account	Both held together: scientuality
Modes of claim (section 3)	Experienced: a report of what it is like; judged by honesty and consistency	Measured: a finding; judged by evidence, replication and effect size	Interpretive: a reading of how things relate; judged by coherence and usefulness
Perspectives of the I	I1, the felt perspective (subjective)	I2, the understanding perspective (objective)	I3, the witness's eye (transjective)

6. An experiential test of I3 (proposed)

An account by someone who had not read the essay (Experienced: another's report; the paper claims nothing from it). From "Eye of the Ship", a poem by Bonnie Antonette, written for the author in 2026,

before the writer had read this essay, and given in full in the Supplementary Material (the Unifying Paper, part 2, A4): "I climbed and climbed, still fearing the below but as I climbed further, my worries began to low. To myself I thought the problems were still there, but then I second guessed as the less I cared. I could've been a giraffe up here as tall as I could climb. But little as a crow I was, sitting with a deeper mind. … The more I looked through the glass, it wasn't other ships but better friends at last. It's not really friends I mean, it's the thoughts that I have, the peace of observation, the space to think at last." The lines are a report, not a result.

The framework has a part that is tried, not measured. The companion essay's four hypotheses (paper II, section 7) can be tested with scales and behavioural measures; I3 is not measured here as a trait, and this section sets out a test any reader can run. Each report is Experienced, judged by honesty. The claim the test bears on, that more readers find a stance from which feeling and understanding are held together at the centre mark than at a matched control, is a claim about the distribution of reports, and would be Measured once run. A positive result would show only that figure and script elicit a stance of the kind the decentering literature describes (section 2), where a matched control does not. It would show nothing about a lens, Π or what awareness is; nor is it Turing's (1950) test: that asks whether talk can pass, this whether a stance is taken up; the Supplementary Material (A3) extends the contrast to artificial systems.

Protocol. Four conditions, each about a minute, each followed by the same written report: (a) felt, (b) understanding, (c) centre and (d) outside, the control. Readers take one of four orders forming a balanced Latin square (a b c d, b d a c, c a d b, d c b a), each condition once in each position and once after each other; position is recorded. The drawing used is Supplementary Figure S13: Figure 1 with every word removed, with two identical marks, a small ring with a point at its centre, one at the centre of the figure and one outside the circle, to its right on the line through the centre, on no edge, seam or corner the figure draws, on two sheets so that on each the mark in use sits at the centre of the page. The figure is an aid to attention, not the object of the test: it gives attention a place, and the eyes then close, since what the test asks about is not seen with them. The script, read aloud or by the reader, is followed exactly:

(a) Felt. "Look at the coloured base of the figure. Close your eyes, and let your attention rest on whatever you feel right now, without naming it or explaining it. Stay with the feeling." (b) Understanding. "Look at the top corner. Close your eyes, and let your attention rest on what you can say about what you feel: what it is, where it comes from, what it means. Stay with the thinking." (c) Centre, on the first sheet, and (d) outside, the control, on the second, with one script for both: "Look at the mark at the centre of the page. Close your eyes, and let your attention rest where the mark was. Notice what is present." The two conditions differ only in which sheet is on the table, so the words tell the reader nothing about what to find.

Report. After each condition the reader answers five prompts: what was present (feeling, thinking, both, neither); if a feeling was present, how near or far it seemed (a five-point scale, very near to very far; otherwise "no feeling present"); whether feeling and thinking could be held at once (yes, partly, no); what the place attention rested on was like, if there was one (empty, full, neither, or a word of your own); and one or two free sentences describing the minute. After each report the reader also completes the Toronto Mindfulness Scale's decentering items, a state measure of that stance (Lau et al. 2006), so that the second prediction has a score for each condition. Two groups: one has only the drawing and the script; the other reads this essay first. Where a group can afford it, some readers are interviewed in Petitmengin's (2006) second-person method, which draws out an experience's structure while managing suggestion, and the interviews compared for recurring features (neurophenomenology; Varela 1996).

Closing: the eye of the witness. After the fourth report, whatever the order, one more minute closes the session. The reader is shown the drawing of Figure 3 without its caption, with this script: "Look at the point at the centre. Close your eyes, and let whatever you feel or think come as large or as small as it comes. Notice whether what is looking grows or shrinks with it." The reader then answers one prompt, whether what was looking changed in size with what it looked at (yes, no,

unsure), and writes one or two free sentences. The step comes last so that nothing in it can tell the reader what to find in the four conditions. It is a demonstration of the witness, the self looking at its own perspectives, not part of the comparison, and no prediction rests on it (Experienced).



Figure 3. The eye of the witness (Interpretive). The triangle in its circle of Figure 1, without words or colour, drawn three times around one point of light. Each circle is the inner circle of the triangle around it, so the drawing could go on inward or outward without end. The frames change size and the point does not. The point is the singularity, Π brought to a point in one person. The rings around it make the eye, I3, and the frames open and close like a camera's aperture, with how present the witness is. Looking at the circles gives a flat eye, one looked at. Looking at the whole gives a further dimension, inward to the point and outward from it: an eye looked through, joining the inner view and the outer view as the edges of Supplementary Figure S12 do. In that reading the figure is transjective (section 2). It is an image, not evidence.

What would count. Three predictions are stated in advance, and a secondary fourth. First, more readers report holding feeling and thinking at once in the centre condition than in the outside condition, "partly" being coded with "no" so that only a clear "yes" counts; if the two do not differ, the effect belongs to resting attention on a mark, not to the centre. The two marks also differ in what surrounds them, the centre mark inside the circle and the triangle and the outside mark in open space, so a positive result may come from the surround rather than the centre; a third sheet, the mark at the centre of a plain circle, would separate the two and is the first follow-up if the result is positive. Second, the decentering score is higher after the centre condition than after the outside condition. Third, both differences appear in the group given only the drawing and the script; if they appear only in readers of the essay, expectation is the better explanation. Fourth, and secondary, more readers describe the centre as empty than the outside mark, in the forced choice or in their free sentences, coded blind. This bears only on the author's report that the centre is felt as hollow (section 5), not on the lens or Π: a success gives the picture the weakest support, since a hollow mark is one mundane explanation; a failure removes it; the main claim is untouched either way. The felt and understanding conditions are reference points, not tests: their scripts direct attention to one side, so fewer reports of holding both are expected there; that difference is not evidence. The planned analysis is a mixed model of each outcome with condition, position and group as predictors and reader as a random effect; the result is reported as proportions and means with their uncertainty, a Measured claim about reports; each report stays testimony.

What would count against. The claim loses its support if, in a run large enough to find the smallest effect of interest, the centre and outside conditions do not differ on either outcome, or if the differences appear only in readers of the essay. That smallest effect is set in advance: a difference of 15 percentage points in "yes" reports, and of 0.4 standard deviations on the decentering score, between the centre and outside conditions. A first run of about eighty readers, ten in each of the

eight cells, is a pilot with no power calculation claimed: a difference of that size or larger would show the predictions holding, and a smaller one would leave the question open rather than answered. A group run needs ethics approval and informed consent, and should be registered first; a reader's own use needs neither, and is a demonstration rather than a test, since every reader of the essay is in the group that has read it. None of this has been run.

7. Objections and replies

- **"The source is introspection and informal observation."** True, and the position statement says so. Introspection can mislead and informal observation is open to bias, which is why the framework is offered to be tested, in section 6 and in the companion essay.
- **"This cannot be tested."** The Interpretive claims are labelled as such and are judged on coherence and usefulness. The experiential test is in section 6 and the measurable parts in the companion essay.
- **"Mixing science and spirit is a category error."** The essay does not ask either to do the other's job. Its labels exist so that a reader always knows which kind of claim is made.
- **"The parallel maps (Freud, Berne, Jung) are loose analogies."** Agreed. They are marked Interpretive and used to illustrate, not to prove; Freud's structural model in particular is dated, which is why contemporary dual-process and empathy research is set beside it in section 2.
- **"The left-brain and right-brain story is discredited." The popular version is, and the essay says so (Nielsen et al. 2013); McGilchrist's claim about styles of attention is different, used here only as a parallel, with its limits stated.**
- **"The I1, I2 and I3 scheme is only a relabelling." In part: it is an organising scheme, and Interpretive, though section 5 says where it is more than that.** What this essay tests itself is the central perspective, through section 6; the scheme's further testable content is in the companion essay: H1 (compassion at the seam of the two empathies), H3 (within-person variation in emphasis) and H4 (indifference and cruelty as different profiles). If the hypotheses all fail, the scheme should be kept, if at all, as a metaphor and not as a framework for arriving at the centre.

Risks of the framework

The framework could be misread in five ways: as presenting spiritual claims as findings, as encouraging people to skip medical or psychological care (nothing in either essay replaces clinical care), as taking the spiral, laser and lens images literally, as one person's view with a stake in it, or as treating arrival at the centre as the removal of the ego or the end of the work, when the state passes, the ego is what acts, and the certainty that comes with the experience is part of the experience rather than a verdict on it. Labels, the position statement and Limits reduce each risk but cannot remove it.

8. Discussion

This essay proposes a transjective framework in which scientific and spiritual accounts of human experience are held together as two views of one thing, provided every descriptive claim declares its mode: Experienced, Measured or Interpretive. Its contribution is a labelling discipline, an arrangement of three perspectives of the I, with the witness's eye, I3, at the centre, a picture of that eye as a lens with the singularity of consciousness at its focal point, and an experiential test each reader can try. The companion essay (paper II) applies it to empathy and compassion, with four hypotheses that can fail: H1, compassion sits at the seam of the two empathies; H2, labelling transfers; H3, within-person variation in emphasis; and H4, indifference and cruelty are different profiles. The later papers carry it further: to paradox (paper III), to the norms families and cultures pass on (paper IV) and to artificial intelligence (paper VI).

The singularity of consciousness: the cosmology the essay carries (Interpretive and Experienced). The cosmology is given in full in the Unifying Paper (part 2, A3). Figure 1 shows a ring at the centre of the perspectives, a point of light within it and a circle through the three corners. The author reads the circle as π, universal consciousness, seen whole; the ring, I3, as the eye through which π passes in one person; and the point as the singularity of consciousness, its focal point: π brought to a point in one person, the aware self of section 5. A life, as it is lived, is that point: life is π brought to a point, the singularity of consciousness; as it is measured, life is the lens, the body that does the focusing; looked through, the same lens is I3, the witness's eye (Interpretive). The self is π emanated, the same in all: all is one and the one are many, each self a fractal of universal consciousness, connected to the others through the singularity (Interpretive: the cosmopsychist reading placed below, named here and not argued). Schrödinger (1944, Epilogue) reached the same point from physics: "consciousness is a singular of which the plural is unknown", the plurality "merely a series of different aspects of this one thing"; the author reads this as a kindred thought, not as evidence. Through the lens of I3 the everything resides: π does not only pass through the eye but is at home in it, as light is at home in a lens that neither makes nor keeps it; so the eye is not the light: I3 is a window, and what passes through it is awareness itself (Interpretive). π is a mark, not a name (the Unifying Paper, part 1, section 3): what it marks is beyond words and thought, and the essay uses the mark so as not to pretend otherwise (Interpretive). Wherever π appears in this essay it is the mark; the number is meant only where a theorem is cited, and the two are never one claim. "The way that can be spoken of is not the constant way; the name that can be named is not the constant name" (Laozi, chapter 1, trans. Lau 1963). The centre is therefore polycentric: every lens has its own focal point, so there are as many centres as persons, and one light focused in all of them; the count is of the lenses the essay can see, and is a lower bound, and whether the one light is focused elsewhere, in the animal or the stone, is asked in the Supplementary Material (the Unifying Paper, part 2, A3) as a question no test answers, and paper VI (section 7) carries it across the range of beings. Arriving at the centre means arriving at the self, the source from which one's own light is given, not at a single centre of everything. The I is a vantage, not an owner: at the centre the vantage falls away, and what remains is awareness itself, with no I standing apart from it (Interpretive); that is the self merged with the all, and the author's report is that one comes back (Experienced). The light at the point is gathered from all directions and given in all; the lens works both ways. From that follows a further reading, the singularity of the all and the many, a polycentric singularity: every lens focusing the one light to its own point, and the lenses joined by the light they exchange, with a value commitment drawn from it; both are in the Supplementary Material (A3, Figures S5 to S11). These are readings of the picture, from the author's own experience: no hypothesis depends on them, the test in section 6 touches them only in its secondary prediction, and the essay still does not say what awareness is.

Where the picture sits (Interpretive). The lens image is a filter picture with relatives in the consciousness literature. James (1898) proposed that the brain may transmit consciousness rather than produce it; Huxley (1954) called the brain a reducing valve; Kelly et al. (2007) revived the transmission view; and Kastrup (2018) argues in the Journal of Consciousness Studies that individual minds are dissociated parts of one universal consciousness. Cosmopsychism makes the metaphysical claim outright: the cosmos is the one fundamental subject, and individual subjects derive from it (Shani 2015; Goff 2017). The image borrows the shape of these views, one light and many lenses, not their claims: it says nothing about how one consciousness becomes many, the combination problem, and is not an account of what consciousness is. The witness itself has an older literature: the witness-consciousness of Advaita Vedānta, defined and defended in the same journal by Albahari (2009) and examined by Fasching (2011), where "witness" names the constant awareness that this essay calls the self, its witness being the act of looking, the words inverted and the structure the same; Deikman's (1996) "'I' = awareness", identifying the sense of I with awareness itself; and Forman's (1998) pure consciousness event. Advaita also has the structure of the centre, in its three bodies and in turīya, the fourth that is not a fourth state; that reading, and a second number, the golden ratio, as an image of the self, are in the Supplementary Material (A3), and nothing here depends on them. The self and the witness of this essay are closer to these than to any filter theory, and I3 taken up closest to decentering (section 2), which is why section 6 tests a stance and not a picture. The picture carries

the labels it does, Interpretive for the reading and Experienced for the report from the centre, because that is all the essay can claim for it; its one Measured contact is section 6's secondary prediction, which bears on a report of emptiness and nothing else.

Limits. The framework comes from one author's introspection and informal observation, and no new data are reported. Most of the essay is Interpretive, and the scheme of perspectives (I1, I2, I3) has measurable content only through the hypotheses of the companion essay (H1, H3 and H4; H2 tests the labelling discipline itself); the experiential test in section 6 is the only way offered here to try I3, and it rests on each reader's own report. The parallels (McGilchrist, Freud, Berne, Jung) are analogies, and the brain basis of the divided-brain story is contested. The religious material, placed in the Supplementary Material, is brief and comparative: two Buddhist images (the Wheel of Life, Supplementary Figure S4, and Indra's net), Advaita's three bodies and turīya, the chakras and third eye of the yogic subtle body, Laozi, Eckhart and the bliss states of A3; it does not claim to represent any tradition's own account, and does not engage theology in depth. The essay does not say what awareness itself is; it works with the aware self, awareness as it appears in a person. The framework and most of the cited evidence come from Western settings and may not generalise to other cultures.

What would change the view. If readers of the script in section 6 report holding feeling and thinking at once no more often at the centre mark than at the control, with no difference on the decentering measure, or only when they have read the essay first, the claim that the centre mark helps a reader take up I3 loses its support. If the hypotheses of the companion essay (H1, H3 and H4) failed, the scheme should be kept, if at all, as a metaphor and not as a framework for arriving at the centre.

Next steps. The experiential test can be tried by any reader now, as a demonstration; as a test it needs readers who have not read the essay, and their reports compared. The measured tests are in the companion essay (paper II, section 7). Others are invited to test, extend or refute any part.

One way to close, in the author's words: "Be careful not to identify with the state you are in, for the achievable transcendental mystery of the centre of all is but a moment of experience away from realising." That is advice, outside the modes; on the reading above, the centre of all is the one light, and the moment is one's own focal point. Section 6 is the way offered here to try it. Science asks how far we can measure; this essay accepts that some of what matters is met by living it, and the two together are more honest than either alone. Measurement comes closer to experience without end and does not arrive, which is the shape of a limit. On this reading, science and the spiritual are not rivals because one is the approach and the other is the point, and the point is only ever reached from the stance that sees the approach whole, which is arriving at the centre (Interpretive). What the framework arrives at can be said in a phrase: the felt is subjective, the understood is objective, and consciousness, brought to a point through the lens of a person, is transjective: the transjectuality of consciousness.

Your centre, our centre, the centre of all.

Declarations

Competing interests. The author founded a community project (named in the cover file) about neighbours sharing tools, skills and help. It is a separate project: it pays the author no salary or other income, and it is not part of the arguments or hypotheses in this essay and does not depend on it.

Funding. No funding, grant or employer support was received for this work.

Ethics. No new data were collected from human participants. No ethics approval was needed.

Permissions. "Eye of the Ship" is quoted in section 6 and reproduced in full in the Supplementary Material (the Unifying Paper, part 2, A4) with the permission of its writer, Bonnie Antonette.

Use of AI. Generative AI (Claude, from Anthropic) was used to help draft and edit the text, for consistency and reference checks, and to render the figures. The model also contributed the design

details of the experiential test in section 6, among them the control condition, the two groups, the report prompts, the use of the Toronto Mindfulness Scale's decentering items, the analysis plan and the pilot of about eighty readers. It also worded the script of the closing step. The author reviewed all content and takes full responsibility for it.

References

Albahari M (2009). Witness-consciousness: its definition, appearance and reality. Journal of Consciousness Studies 16(1):62-84.

Atmanspacher H, Rickles D (2022). Dual-Aspect Monism and the Deep Structure of Meaning. Routledge.

Barbour IG (1997). Religion and Science: Historical and Contemporary Issues. HarperSanFrancisco.

Berne E (1964). Games People Play: The Psychology of Human Relationships. Grove Press.

Bernstein A, Hadash Y, Lichtash Y, Tanay G, Shepherd K, Fresco DM (2015). Decentering and related constructs: a critical review and metacognitive processes model. Perspectives on Psychological Science 10(5):599-617. https://doi.org/10.1177/1745691615594577

Dahl CJ, Lutz A, Davidson RJ (2015). Reconstructing and deconstructing the self: cognitive mechanisms in meditation practice. Trends in Cognitive Sciences 19(9):515-523. https://doi.org/10.1016/j.tics.2015.07.001

Decety J, Jackson PL (2004). The functional architecture of human empathy. Behavioral and Cognitive Neuroscience Reviews 3(2):71-100. https://doi.org/10.1177/1534582304267187

Deikman AJ (1996). 'I' = awareness. Journal of Consciousness Studies 3(4):350-356.

Evans JStBT, Stanovich KE (2013). Dual-process theories of higher cognition: advancing the debate. Perspectives on Psychological Science 8(3):223-241. https://doi.org/10.1177/1745691612460685

Fasching W (2011). 'I am of the nature of seeing': phenomenological reflections on the Indian notion of witness-consciousness. In Siderits M, Thompson E, Zahavi D (eds), Self, No Self? Perspectives from Analytical, Phenomenological, and Indian Traditions, pp. 193-216. Oxford University Press.

Forman RKC (1998). What does mysticism have to teach us about consciousness? Journal of Consciousness Studies 5(2):185-201.

Freud S (1923). The Ego and the Id. In Strachey J (ed. and trans.), The Standard Edition of the Complete Psychological Works of Sigmund Freud, vol. 19. Hogarth Press and the Institute of Psycho-Analysis (1961).

Gibbins I (2010). Hemispheric wars. Review of The Master and His Emissary: The Divided Brain and the Making of the Western World, by Iain McGilchrist. Australian Book Review 323: 59-60.

Goff P (2017). Consciousness and Fundamental Reality. Oxford University Press.

Gould SJ (1997). Nonoverlapping magisteria. Natural History 106(2):16-22.

Hayes SC, Luoma JB, Bond FW, Masuda A, Lillis J (2006). Acceptance and commitment therapy: model, processes and outcomes. Behaviour Research and Therapy 44(1):1-25. https://doi.org/10.1016/j.brat.2005.06.006

Huxley A (1954). The Doors of Perception. Chatto and Windus.

James W (1898). Human Immortality: Two Supposed Objections to the Doctrine. Houghton, Mifflin.

Josipovic Z (2014). Neural correlates of nondual awareness in meditation. Annals of the New York Academy of Sciences 1307:9-18. https://doi.org/10.1111/nyas.12261

Jung CG (1966). Two Essays on Analytical Psychology. Collected Works, vol. 7. Princeton University Press.

Kastrup B (2018). The universe in consciousness. Journal of Consciousness Studies 25(5-6):125-155.

Kelly EF, Kelly EW, Crabtree A, Gauld A, Grosso M, Greyson B (2007). Irreducible Mind: Toward a Psychology for the 21st Century. Rowman and Littlefield.

Laozi (c. 4th century BCE). Tao Te Ching. Translated by D.C. Lau (1963). Penguin.

Lau MA, Bishop SR, Segal ZV, Buis T, Anderson ND, Carlson L, Shapiro S, Carmody J, Abbey S, Devins G (2006). The Toronto Mindfulness Scale: development and validation. Journal of Clinical Psychology 62(12):1445-1467. https://doi.org/10.1002/jclp.20326

McGilchrist I (2009). The Master and His Emissary: The Divided Brain and the Making of the Western World. Yale University Press.

Metzinger T (2020). Minimal phenomenal experience: meditation, tonic alertness, and the phenomenology of "pure" consciousness. Philosophy and the Mind Sciences 1(I):7. https://doi.org/10.33735/phimisci.2020.I.46

Nielsen JA, Zielinski BA, Ferguson MA, Lainhart JE, Anderson JS (2013). An evaluation of the left-brain vs. right-brain hypothesis with resting state functional connectivity MRI. PLoS ONE 8(8):e71275. https://doi.org/10.1371/journal.pone.0071275

Peters S (2012). The Chimp Paradox. Vermilion.

Petitmengin C (2006). Describing one's subjective experience in the second person: an interview method for the science of consciousness. Phenomenology and the Cognitive Sciences 5(3-4):229-269. https://doi.org/10.1007/s11097-006-9022-2

Preston SD, de Waal FBM (2002). Empathy: its ultimate and proximate bases. Behavioral and Brain Sciences 25(1):1-20. https://doi.org/10.1017/S0140525X02000018

Schooler JW (2002). Re-representing consciousness: dissociations between experience and meta-consciousness. Trends in Cognitive Sciences 6(8):339-344. https://doi.org/10.1016/S1364-6613(02)01949-6

Schrödinger E (1944). What Is Life? The Physical Aspect of the Living Cell. Cambridge University Press. Epilogue, "On Determinism and Free Will" (reprinted in What Is Life? with Mind and Matter and Autobiographical Sketches, Cambridge University Press, 1992).

Shamay-Tsoory SG, Aharon-Peretz J, Perry D (2009). Two systems for empathy: a double dissociation between emotional and cognitive empathy in inferior frontal gyrus versus ventromedial prefrontal lesions. Brain 132:617-627. https://doi.org/10.1093/brain/awn279

Shani I (2015). Cosmopsychism: a holistic approach to the metaphysics of experience. Philosophical Papers 44(3):389-437.

Singer T, Seymour B, O'Doherty J, Kaube H, Dolan RJ, Frith CD (2004). Empathy for pain involves the affective but not sensory components of pain. Science 303:1157-1162. https://doi.org/10.1126/science.1093535

Smith A (1759). The Theory of Moral Sentiments. A. Millar, London, and A. Kincaid and J. Bell, Edinburgh.

Solms M, Panksepp J (2012). The "Id" knows more than the "Ego" admits: neuropsychoanalytic and primal consciousness perspectives on the interface between affective and cognitive neuroscience. Brain Sciences 2(2):147-175. https://doi.org/10.3390/brainsci2020147

Turing AM (1950). Computing machinery and intelligence. Mind 59(236):433-460. https://doi.org/10.1093/mind/LIX.236.433

Varela FJ (1996). Neurophenomenology: a methodological remedy for the hard problem. Journal of Consciousness Studies 3(4):330-349.

Varela FJ, Shear J (1999). First-person methodologies: what, why, how? Journal of Consciousness Studies 6(2-3):1-14.

Vervaeke J (2019). Ep. 31 - Awakening from the Meaning Crisis - Embodied-Embedded RR as Dynamical-Developmental GI [video]. YouTube, 16 August 2019. https://www.youtube.com/watch?v=gfKcVbNd7Xc

Zahavi D (2005). Subjectivity and Selfhood: Investigating the First-Person Perspective. MIT Press.

Zaki J, Ochsner KN (2012). The neuroscience of empathy: progress, pitfalls and promise. Nature Neuroscience 15:675-680. https://doi.org/10.1038/nn.3085